Feature Extraction Techniques

A Machine Learning Approach

What is Feature Extraction?

- Feature **extraction** creates a new feature from the existing features.
- Purpose is to reduce a dataset dimensionality
- Most popular feature extraction techniques:
  - Principal Component Analysis (PCA)
  - Linear Discriminant Analysis (LDA) – self study
- Datasets may have hundreds (or even thousands) of features.
- If the number of features becomes similar or even bigger!, then this can most likely lead to overfitting.

Advantages of feature extraction

Using Regularization could certainly help reduce the risk of overfitting, but using Feature Extraction techniques can also lead to other types of advantages such as:

- Accuracy improvements.
- Overfitting risk reduction.
- Speed up in training.
- Improved Data Visualization.
- Increase in understanding of our model.

Dimension Reduction using Principal Component Analysis (PCA)

PCA is …

A useful statistical technique that has found application in fields such as face recognition and image compression, and is a common technique for finding patterns in data of high dimension.

PCA Steps

- Mean
- Covariance
- Eigenvalues, Eigenvectors and Normalized Eigenvectors
- Derive the Dataset
- Plot the points

PCA – Example

- Use PCA to reduce the dimension from 2 to 1 for the following data

Feature	X	y
Dp-1	4	11
Dp-2	8	4
Dp-3	13	5
Dp-4	7	14

1. Find number of features and samples:

- Number of features, n = 2
- Number of samples, N = 4

2. Find the mean of two variables x and y

$$\bar{x} = \frac{4 + 8 + 13 + 7}{4} = 8$$

$$\bar{y} = \frac{11 + 4 + 5 + 14}{4} = 8.5$$

Feature	X	y
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Computation of covariance matrix

- Ordered pairs: (x, x), (x, y), (y, x) and (y, y)
- Covariance of all ordered pairs

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$$Cov(x, x) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j)$$

Computation of covariance matrix

- Ordered pairs: (x, x), (x, y), (y, x) and (y, y)
- Covariance of all ordered pairs

$$cov(x,x) = \frac{1}{4-1}\left[(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2\right]$$

$$cov(x,x) = 14$$

Feature	X	y
Dp-1	4	11
Dp-2	8	4
Dp-3	13	5
Dp-4	7	14

$$Cov(x,y) = \frac{1}{4-1}\left[(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)\right]$$

$$= -11$$

$$cov(y,x) = -11$$

$$cov(y,y) = \frac{1}{4-1}\left[(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2\right]$$

$$= 23$$

Feature	X	y
Dp-1	4	11
Dp-2	8	4
Dp-3	13	5
Dp-4	7	14

Covariance matrix

$$S = \begin{bmatrix} cov(x,x) & cov(x,y) \\ cov(y,x) & cov(y,y) \end{bmatrix}$$

$$= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Find the Eigen value and Eigen vectors and Normalized Eigen vectors

Eigen value :

$$|(S-\lambda I)| = 0 \quad \Rightarrow \quad \left| \begin{bmatrix} (14-\lambda) & -11 \\ -11 & (23-\lambda) \end{bmatrix} \right| = 0$$

$$\begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\left| \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \right| = 0$$

$$(14-\lambda)(23-\lambda) - (-11 \times -11) = 0$$

$$\lambda^2 - 37\lambda + 201 = 0$$

Solving for λ using

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = 30.385 \quad \text{or} \quad 6.62$$

$\lambda = 30.3849,\ 6.6151$

$\lambda_1 = 30.3849\ ;\ \lambda_2 = 6.6151$

$\lambda_1 > \lambda_2$

λ_1 is First PC

(ii) Eigen vector of λ_1

$$(S - \lambda_1 I) U_1 = 0$$

$$\begin{bmatrix} (14 - \lambda_1) & -11 \\ -11 & (23 - \lambda_1) \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} (14-\lambda_1)u_1 - 11u_2 \\ -11u_1 + (23-\lambda_1)u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(14-\lambda_1)u_1 - 11u_2 = 0$$

$$-11u_1 + (23-\lambda_1)u_2 = 0$$

$$(14-\lambda_1)u_1 = 11u_2$$

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda_1} = t$$

when $t = 1 \Rightarrow u_1 = 11 ; u_2 = 14-\lambda_1$

Eigen vector u_1 of $\lambda_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix}$

$$= \begin{bmatrix} 11 \\ 14 - 30.3849 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix}$$

Normalize the eigen vector U_1

$$e_1 = \begin{bmatrix} \frac{11}{\sqrt{11^2 + (-16.3849)^2}} \\ \frac{-16.3849}{\sqrt{11^2 + (-16.3849)^2}} \end{bmatrix} = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

Derive New Dataset :

	Ex_1	Ex_2	Ex_3	Ex_4
FPC (PC1)	P_{11}	P_{12}	P_{13}	P_{14}

$$P_{11} = e_1^T \begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} -4 \\ 2.5 \end{bmatrix} = -4.3052$$

$$P_{12} = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} 8-8 \\ 4-8.5 \end{bmatrix}$$

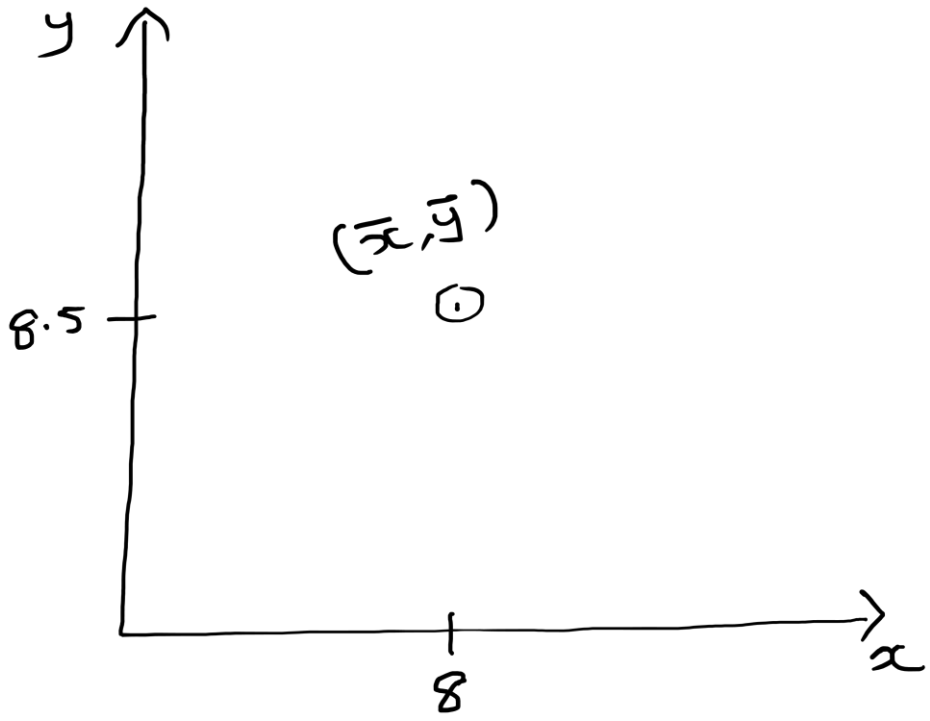
$$= 3.7361$$

$$P_{13} = 5.6928$$

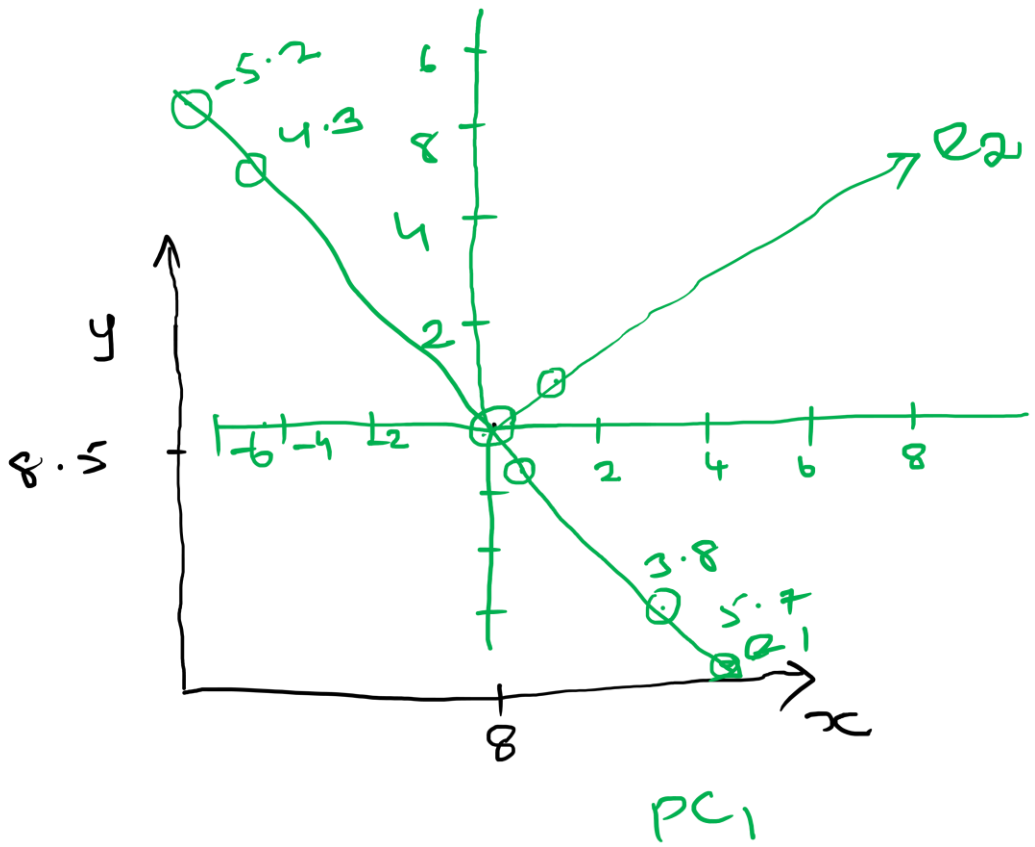
$$P_{14} = -5.1238$$

	Ex1	Ex2	Ex3	Ex4
PC1	-4.3052	3.7361	5.6928	-5.1238

Coordinate system for principal component



$$e_1 = \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix} = \begin{bmatrix} 0.557 \\ -0.84 \end{bmatrix}$$



PC1

Thank you